AQ6.2: Activity Questions 2 - Not Graded

This assignment will not be graded and is only for practice.

Level 1

1 point

Choose the set of correct options.

- [ ] If v_1 v1 and v_2 v2 are in the kernel of a linear transformation $T: V \rightarrow W$ T : V → W, where V V and W W are two vector spaces, then $v_1 + v_2$ v1 + v2 will also be in the kernel of T T.
- [ ] Kernel of a linear transformation $T: V \rightarrow W$ T : V → W is a vector subspace of V V, where V V and W W are two vector spaces.
- [ ] If w_1 w1 and w_2 w2 are in the image of a linear transformation $T: V \rightarrow W$ T : V → W, where V V and W W are two vector spaces, then $w_1 + w_2$ w1 + w2 will also be in the image of T T.
- [ ] Image of a linear transformation $T: V \rightarrow W$ T : V → W is a vector subspace of W W, where V V and W W are two vector spaces.

No, the answer is incorrect.

Score: 0

Accepted Answers:

If v_1 v1 and v_2 v2 are in the kernel of a linear transformation $T: V \rightarrow W$ T : V → W, where V V and W W are two vector spaces, then $v_1 + v_2$ v1 + v2 will also be in the kernel of T T.

Kernel of a linear transformation $T: V \rightarrow W$ T : V → W is a vector subspace of V V, where V V and W W are two vector spaces.

If w_1 w1 and w_2 w2 are in the image of a linear transformation $T: V \rightarrow W$ T : V → W, where V V and W W are two vector spaces, then $w_1 + w_2$ w1 + w2 will also be in the image of T T.

Image of a linear transformation $T: V \rightarrow W$ T : V → W is a vector subspace of W W, where V V and W W are two vector spaces.

Consider the following linear transformation:

$T: R^3 \rightarrow R^2$ T : R3 → R2

$T(x, y, z) = (2x + 3z, 4y + z)$ T (x, y, z) = (2x + 3z, 4y + z)

Answer questions 2,3,4 and 5, using the information given above.

1 point

Which of the following matrices corresponds to the given linear transformation T T with respect to the standard ordered basis for R^3 R3 and the standard ordered basis for R^2 R2 ?

- ○ $\begin{bmatrix} 2 & 3 & 0 \\ 4 & 1 & 0 \end{bmatrix}$ [230410] [243100]
- ○ $\begin{bmatrix} 2 & 0 \\ 0 & 4 \\ 3 & 1 \end{bmatrix}$ [20043131] [203041]
- ○ $\begin{bmatrix} 2 & 0 & 3 \\ 0 & 4 & 1 \end{bmatrix}$ [203041] [200431]

○ $\begin{bmatrix} 24 \\ 00 \end{bmatrix}$ $\begin{bmatrix} 31 \\ 23 \end{bmatrix}$ $\begin{bmatrix} 04 \\ 10 \end{bmatrix}$

No, the answer is incorrect.
Score: 0
Accepted Answers:
$\begin{bmatrix} 203 \\ 041 \end{bmatrix}$ $\begin{bmatrix} 20 \\ 04 \\ 31 \end{bmatrix}$

1 point
Which of the following represents a basis of the kernel of TT?

○ $\{(-\frac{3}{2}, 0, 1), (0, -\frac{1}{4}, 1)\}\{(-23, 0, 1), (0, -41, 1)\}$

○ $\{(-\frac{3}{2}, -\frac{1}{4}, 1)\}\{(-23, -41, 1)\}$

○ $\{(-\frac{3}{2}, -\frac{1}{4}, 2)\}\{(-23, -41, 2)\}$

○ $\{(2, 0, 3), (0, 4, 1)\}\{(2, 0, 3), (0, 4, 1)\}$

No, the answer is incorrect.
Score: 0
Accepted Answers:
$\{(-\frac{3}{2}, -\frac{1}{4}, 1)\}\{(-23, -41, 1)\}$

What will be the dimension of the subspace $Im(T)$Im(T)?

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 2
1 point
1 point
Choose the correct option.

○ TT is an isomorphism.

○ TT is one to one but not onto.

○ TT is onto but not one to one.

○ TT is neither one to one, nor onto.

No, the answer is incorrect.
Score: 0
Accepted Answers:
TT is onto but not one to one.

1 point
Choose the set of correct options.

☐ There exists a linear transformation $T: R^2 \rightarrow R^2$T : R2 → R2 whose range is contained in the circle $x^2 + y^2$ = 1x2 + y2 = 1.

☐ There exists a linear transformation $T: R^2 \rightarrow R^2$ T : R2 → R2 whose range equals the line $x = y$ x = y.

☐ There exists a linear transformation $T: R^2 \rightarrow R^2$ T : R2 → R2 whose range is contained in the set $S = \{(x, y) | x > 0\}$ S = {(x, y) | x > 0}.

☐ There exists a linear transformation $T: R^2 \rightarrow R^2$ T : R2 → R2 whose range equals the line $y = \pi x$ y = πx.

No, the answer is incorrect.

Score: 0

Accepted Answers:

There exists a linear transformation $T: R^2 \rightarrow R^2$ T : R2 → R2 whose range equals the line $x = y$ x = y.

There exists a linear transformation $T: R^2 \rightarrow R^2$ T : R2 → R2 whose range equals the line $y = \pi x$ y = πx.

Level 2

1 point

Which option represents the kernel and image of the following linear transformation?

$T: R^2 \rightarrow R^2$ T : R2 → R2
$T(x, y) = (x, 0)$ T(x, y) = (x, 0)

○ $ker(T) = Span\{(1, 0)\}, Im(T) = Span\{(1, 0)\}$ ker(T) = Span{(1, 0)}, Im(T) = Span{(1, 0)}

○ $ker(T) = Span\{(1, 0)\}, Im(T) = Span\{(0, 1)\}$ ker(T) = Span{(1, 0)}, Im(T) = Span{(0, 1)}

○ $ker(T) = Span\{(0, 1)\}, Im(T) = Span\{(1, 0)\}$ ker(T) = Span{(0, 1)}, Im(T) = Span{(1, 0)}

○ $ker(T) = Span\{(0, 1)\}, Im(T) = Span\{(0, 1)\}$ ker(T) = Span{(0, 1)}, Im(T) = Span{(0, 1)}

No, the answer is incorrect.

Score: 0

Accepted Answers:

$ker(T) = Span\{(0, 1)\}, Im(T) = Span\{(1, 0)\}$ ker(T) = Span{(0, 1)}, Im(T) = Span{(1, 0)}

1 point

Which of the following linear transformations is an isomorphism between the vector spaces $V = \{(x, y, z) \mid x = y - z, \text{ and } x, y, z \in R\} \subset R^3$ V = {(x, y, z) | x = y − z, and x, y, z ∈ R} ⊂ R3 and $W = \{(x, y, z) \mid x = y, \text{ and } x, y, z \in R\} \subset R^3$ W = {(x, y, z) | x = y, and x, y, z ∈ R} ⊂ R3?

○ $T: V \rightarrow W$ T : V → W, such that $T(x, y, z) = (y, y, 0)$ T(x, y, z) = (y, y, 0)

○ $T: V \rightarrow W$ T : V → W, such that $T(x, y, z) = (y, y, z)$ T(x, y, z) = (y, y, z)

○ $T: V \rightarrow W$ T : V → W, such that $T(x, y, z) = (x, y, z)$ T(x, y, z) = (x, y, z)

○ There does not exist any isomorphism between V V and W W.

No, the answer is incorrect.

Score: 0

Accepted Answers:
$T: V \rightarrow W$T : V → W, such that $T(x, y, z) = (y, y, z)$T(x, y, z) = (y, y, z)

1 point

Which of the following linear transformations is an isomorphism between the vector spaces $V = \{(x, y, z) \mid x = y - z = 0, \text{ and } x, y, z \in R\} \subset R^3$V = {(x, y, z) | x = y − z = 0, and x, y, z ∈ R} ⊂ R3 and $W = \{(x, y, z) \mid x = y = 0, \text{ and } x, y, z \in R\} \subset R^3$W = {(x, y, z) | x = y = 0, and x, y, z ∈ R} ⊂ R3?

- $T: V \rightarrow W$T : V → W, such that $T(x, y, z) = (x, 0, y)$T(x, y, z) = (x, 0, y).
- $T: V \rightarrow W$T : V → W, such that $T(x, y, z) = (0, y, z)$T(x, y, z) = (0, y, z).
- $T: V \rightarrow W$T : V → W, such that $T(x, y, z) = (0, 0, x)$T(x, y, z) = (0, 0, x).
- There does not exist any isomorphism between VV and WW.

No, the answer is incorrect.
Score: 0

Accepted Answers:
$T: V \rightarrow W$T : V → W, such that $T(x, y, z) = (x, 0, y)$T(x, y, z) = (x, 0, y).

1 point

Consider the linear transformation $T: R^3 \rightarrow R^3$T : R3 → R3 such that $T(x, y, z) = (x - y + z, 2x + y - z, -x - 2y + 2z)$T(x, y, z) = (x − y + z, 2x + y − z, −x − 2y + 2z).

Which of the following options are true about TT?

- Range of TT is $\{(x, y, z) \in R^3 \mid x = y + z\}${(x, y, z) ∈ R3 | x = y + z}.
- Range of TT is $\{(x, y, z) \in R^3 \mid x = 0, y + z = 0\}${(x, y, z) ∈ R3 | x = 0, y + z = 0}.
- Rank of TT is 22
- Rank of TT is 11

No, the answer is incorrect.
Score: 0

Accepted Answers:
Range of TT is $\{(x, y, z) \in R^3 \mid x = y + z\}${(x, y, z) ∈ R3 | x = y + z}.
Rank of TT is 22

Check Answers
Your score is: 0/10